

Task 3: Attention Visualization & Token Attribution

TinyLlama Instruction Tuning Project

March 27, 2026

Task description

This analytical phase, "Attention Visualization & Token Attribution," investigates how Low-Rank Adaptation (LoRA) mathematically shifts the internal attention mechanisms of TinyLlama. We loaded a fine-tuned LoRA checkpoint (rank 16, train on 10k Alpaca examples) and evaluated the structural routing of all 704 attention heads ($32 \text{ heads} \times 22 \text{ layers}$) across 20 distinct instructional prompts. The task encapsulates multiple sub-objectives: extracting deep attention maps, implementing cosine-similarity based feature clustering to uncover semantic head groupings, mapping token attribution flows using attention rollout matrices, and generating interpretable HTML heatmaps. By empirically classifying heads as "instruction-following" or "completion-centric", we isolate and quantify the exact architectural alterations triggered by fine-tuning.

Procedure & technical implementation

1. **Attention Map Extraction:** We isolated the base TinyLlama-1.1B-Chat-v1.0 model and merged it with the Alpaca LoRA adapter. We monkey-patched the transformer `self_attn.__call__` logic specifically over a forced sequence forward-pass to capture the raw softmax matrices (H, L, L) per layer simultaneously across the 20 Alpaca inference samples.
2. **Token Attribution (Rollout):** We simulated token influence flows from embedded input queries to generated output text. This was achieved by determining the cumulative product $R^{(l)} = \text{norm}(A^{(l)} + I) \cdot R^{(l-1)}$ incrementally down the model's 22 nested layer blocks.
3. **Clustering Analysis:** To group head behavior, we constructed a 40-dimensional feature profile for each head by concatenating mean instruction and mean completion attention scores spanning the 20 test inputs. We mapped groups utilizing average-linkage hierarchical clustering over spatial cosine distances with $k = 3$ distinct clusters.
4. **Head Function Classification:** We established an algorithmic classification threshold per head. If the mean attention mass directed toward prompt tokens exceeded the local-window response attention, the head was tagged as **instruction-following**. Conversely, if attention leaned toward recent generation context, it was tagged as **completion**.
5. **Visualization Exports:** Comprehensive comparative heatmaps scaling the base vs LoRA distributions were assembled. The pipeline outputted 20 distinct topological HTML attention graphs alongside Python clustering artifacts natively stored within `results/task3/`.

Metrics Calculation Methodology

To ensure reproducibility, performance shifts within internal weights were tracked via the following metrics:

- **Instruction-Following vs. Completion classification:** A binary toggle determined per-head. Calculated mathematically by taking the average attention weight addressing the "prompt" token indices and subtracting the average weight traversing the latest "completion" token sequence window.
- **Attention Focus:** A measurement of graph sharpness per evaluated sequence. Calculated by extracting the maximum attention weight scalar applied to any single target token by a head.
- **Attention Entropy:** Derived utilizing standard Shannon information theory over the softmax probability vectors mapping token correlation. A lower entropy figure inherently signifies a highly concentrated and deterministic focal target.
- **Δ Net (Layer Level):** Measured comprehensively across layer clusters by subtracting the net instruction-completion orientation of the target baseline model from the identical structure within the LoRA modification.

Results

Table 1: Head classification and attention statistics (20 Alpaca prompts, 704 heads). Source: `analysis_summary.json`.

Model	Instruction-following	Completion	Attn. focus	Entropy
Base	445 (63.2%)	259 (36.8%)	0.5445	0.7355
LoRA	417 (59.2%)	287 (40.8%)	0.5999	0.7289
Δ	-28 heads (-4.0%)	+28	+10.2%	-0.9%

Key findings.

- **28 heads (4%)** shifted instruction→completion after LoRA; all 22 layer Δ_{net} values are negative, confirming a uniform completion-mode amplification across depth.
- **Attention focus +10.2%** (0.5445→0.5999) with slightly lower entropy: LoRA sharpens prompt-token attribution without dispersing attention.
- **Most affected layers:** 16 ($\Delta=-0.00301$), 12 (-0.00248), 13 (-0.00243), 9 (-0.00236), 2 (-0.00208) — mid-to-late layers dominate LoRA’s recalibration.
- **Stable instruction heads:** L8H11, L15H16, L11H8, L15H20 retain near-identical scores in both models; L1H30 and L14H30 completion heads are amplified in magnitude by LoRA.
- **Cluster structure preserved:** 3-group dendrogram topology identical in base and LoRA; LoRA makes within-cluster refinements only.

Evidence (screenshots)

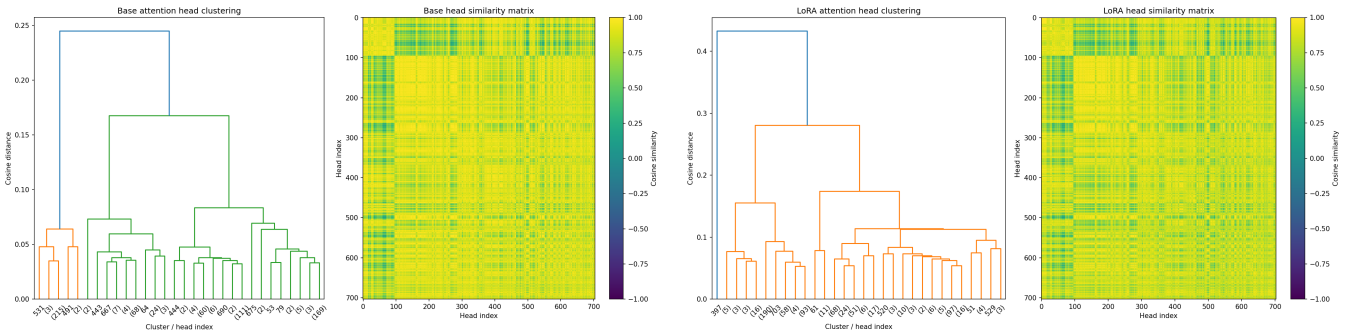


Figure 1: Attention head clustering dendrogram topologies mapping base unsteered (left) vs Alpaca LoRA-tuned (right).

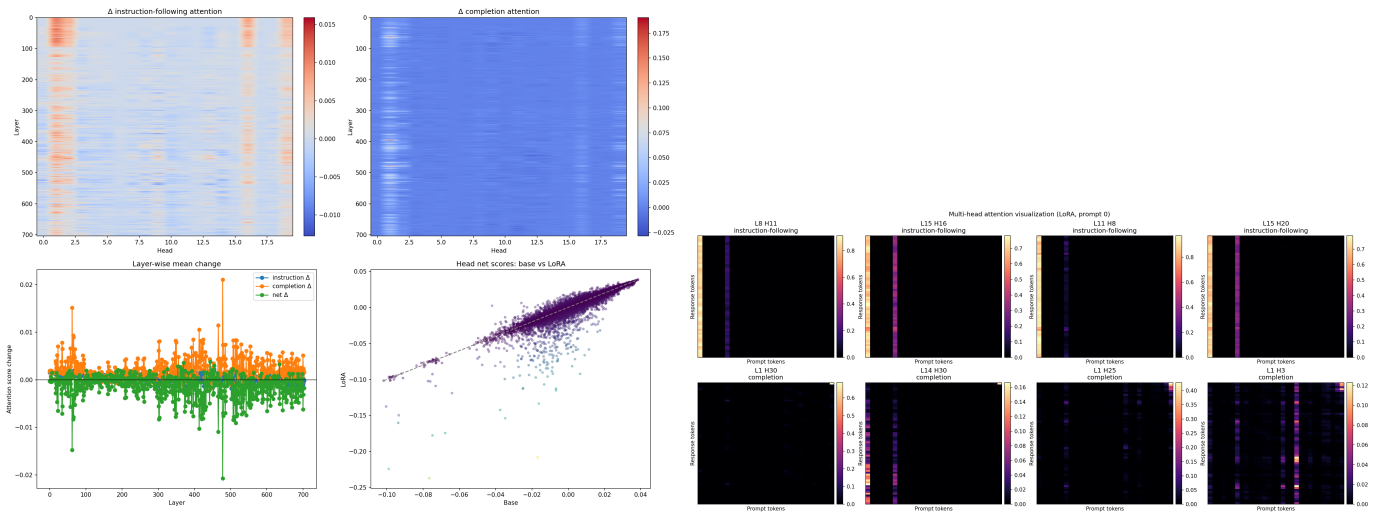


Figure 2: **Left:** Scatter highlighting broad net-completion shift orientation. **Right:** Multi-head attention instruction vs. completion contrast graph mapping specific prompt tokens.